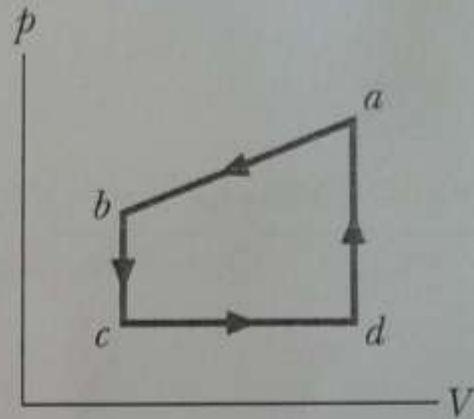


- (b) The figure shows a closed cycle for a gas (not drawn to scale). The change in the internal energy of the gas as it moves from a to c along the path abc is -200 J . As it moves from c to d , 180 J must be transferred to it as heat. An additional transfer of 80 J in the form of heat is needed as it moves from d to a .



Die figuur dui 'n geslote siklus vir 'n gas (nie op skaal geteken nie). Die verandering in interne energie van die gas wanneer dit van a na c toe beweeg langs die pad abc is -200 J . Wanneer dit van c na d beweeg moet 180 J oorgedra word in die vorm van warmte. 'n Addisionele 80 J in die vorm van warmte word dan verder benodig om van d na a te beweeg. Hoeveel arbeid word verrig deur die gas wanneer die sisteem van c na d beweeg?

[4]

$$\Delta E_{abc} = -200\text{ J}$$

$$\Delta E_{cda} = +200\text{ J} \checkmark$$

Because in a cyclic system $\Delta E_{int} = 0\text{ J} \checkmark$

Work done from d to a is zero (no volume change)

$$\Delta E_{cda} = Q - W = 200\text{ J} \checkmark$$

$$= Q_{cd} + Q_{da} - (W_{cd} + W_{da})$$

$$W_{cd} = Q_{cd} + Q_{da} - \Delta E_{cda}$$

$$= 180 + 80 - 200$$

$$= 60\text{ J} \checkmark$$

- (c) What mass of ice at its melting point, must be mixed with 50 g of steam at 100°C , in a thermally insulated container, to produce liquid water at 40.0°C ?

Water massa ys by dié se smelt punt moet met 50 g stoom by 100°C gemeng word in 'n termies geïsoleerde houer, om vloeistof water by 40.0°C te kry?

$$c_{\text{water}} = 4180\text{ J/kg}$$

$$L_{f,\text{water}} = 333\text{ kJ/kg}$$

$$L_{v,\text{water}} = 2256\text{ kJ/kg}$$

[6]

Let the mass of the ice be m_i and that of the steam be m_s . Then

$$\Delta Q = 0 = Q_{\text{ice/water}} + Q_{\text{steam/water}}$$

$$Q_{\text{ice/water}} = -Q_{\text{steam/water}}$$

$$L_f m_i + c_w m_i (T_f - 0.0^\circ\text{C}) = m_s L_v - m_s c_w (T_f - 100^\circ\text{C})$$

where $T_f = 40^\circ\text{C}$ is the final temperature. We solve for m_i :

$$L_f m_i + c_w m_i (T_f - T_{i,\text{ice}}) = m_s L_v - m_s c_w (T_f - T_{i,\text{steam}})$$

$$m_i = \frac{-L_v m_s - c_w m_s (T_f - 100.0^\circ\text{C})}{L_f + c_w (T_f - 0^\circ\text{C})}$$

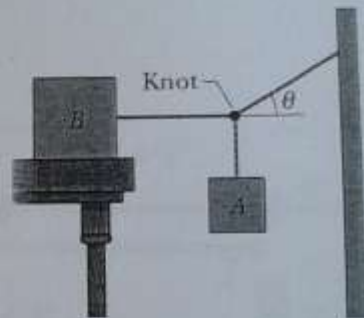
$$= \frac{(2256 \times 10^3\text{ J/kg})(50 \times 10^{-3}\text{ kg}) + (4180\text{ J/kg.K})(50 \times 10^{-3}\text{ kg})(40^\circ\text{C} - 100.0^\circ\text{C})}{(333 \times 10^3\text{ J/kg}) - (4180\text{ J/kg.K})(40^\circ\text{C} - 0^\circ\text{C})}$$

$$= 250 \times 10^{-3}\text{ kg} \approx 250.0\text{ g}$$

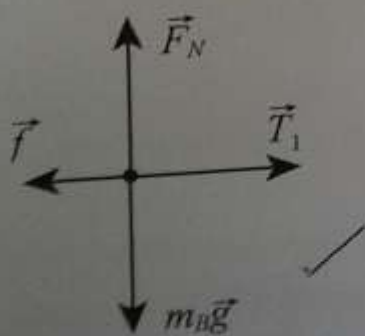
Question 3 / Vraag 3

Block B in the figure weighs 711 N. The coefficient of static friction between the block and table is 0.25; the angle θ is 30° . Assume that the cord between B and the knot is horizontal. Find the maximum weight of block A for which the system will be stationary.

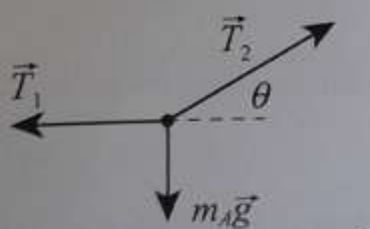
Die gewig van blok B in die figuur is 711 N. Die statiese wrywingskoëffisiënt tussen die blok en die tafel is 0.25. Die hoek θ wat aangedui is, is 30° . Veronderstel dat die tou tussen B en die knoop horisontaal is. Bereken die maksimum gewig van blok A, sodat die stelsel nie beweeg nie.



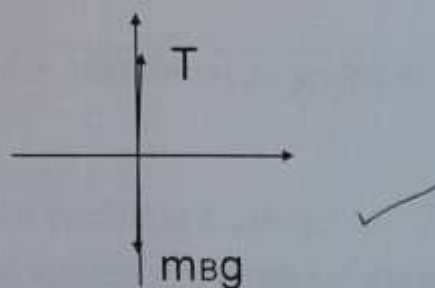
[3+9]



Block B



The knot



Block A

$\vec{T}_1$ is the tension force of the rope pulling on block B or pulling on the knot (as the case may be), $\vec{T}_2$ is the tension force exerted by the second rope (at angle $\theta = 30^\circ$) on the knot, $\vec{f}$ is the force of static friction exerted by the horizontal surface on block B, $\vec{F}_N$ is normal force exerted by the surface on block B, W_A is the weight of block A (W_A is the magnitude of $m_A \vec{g}$), and W_B is the weight of block B ($W_B = 711 \text{ N}$ is the magnitude of $m_B \vec{g}$).

Applying NII law to block B and knot:

Block B

$$\sum F_{Bx} = T_1 - f_s = m_B a_x = 0 \quad \checkmark$$

$$\sum F_{By} = N_B - m_B g = m_B a_y = 0 \quad \checkmark$$

$$T_1 = f_s = \mu_s N_B$$

$$N_B = m_B g$$

Then:

$$T_1 = \mu_s N_B = \mu_s m_B g$$

$$= 178 \text{ N}$$

the knot:

$$\sum F_{\text{knot},x} = -T_1 + T_2 \cos \theta = m_{\text{knot}} a_x = 0 \quad \checkmark$$

$$\sum F_{\text{knot},y} = T_2 \sin \theta - m_A g = m_{\text{knot}} a_y = 0 \quad \checkmark$$

$$-T_1 + T_2 \cos \theta = 0$$

$$T_2 = \frac{T_1}{\cos \theta} = 205 \cdot 3 \text{ N}$$

Substituting:

$$W_A = m_A g = T_2 \sin \theta = \frac{T_1 \sin \theta}{\cos \theta}$$

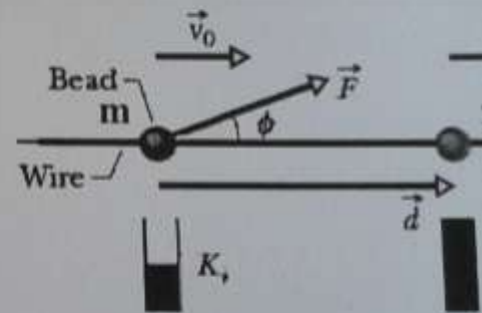
$$= \frac{\mu_s m_B g \sin \theta}{\cos \theta}$$

$$= 102.6 = 103 \text{ N}$$

$$\approx 1.0 \times 10^2 \text{ N} \quad \checkmark$$

Question 4 / Vraag 4

- (a) Consider a bead which can slide on a frictionless wire, as indicated in the figure. A constant force $\vec{F}$ is applied at an angle ϕ to the wire and causes the bead to accelerate. Using this scenario, derive the Kinetic Energy-Work theorem, $\Delta K = W_{\text{net}}$, showing clearly all your assumptions and reasoning.



Beskou 'n kraletjie wat langs 'n wrywinglose draad gly, soos aangedui in die figuur. 'n Konstante krag $\vec{F}$ word toegepas teen 'n hoek ϕ met die draad, en dit veroorsaak dat die kraletjie versnel. Gebruik hierdie situasie, en herlei die Kinetiese Energie-Arbeid Stelling, $\Delta K = W_{\text{net}}$. Gee al jou reëls en aannames duidelik in jou antwoord.

Since the acceleration is only along the wire - x- direction, only the x-component of the force will contribute to the acceleration.

$$F \cos \phi = F_x = ma_x \quad (1) \quad \checkmark$$

The bead then moves through a displacement $\vec{d}$, which is parallel to the wire. $\checkmark$

Starting with an initial v_{0x} the bead accelerates to a final velocity v , through a distance $|\vec{d}| = \Delta x$

From equations of motion it is known that:

$$v_f^2 = v_{0x}^2 + 2a\Delta x \quad \checkmark$$

Then

$$a = \frac{v_f^2 - v_{0x}^2}{2\Delta x} \quad \checkmark$$

Substituting into (1)

$$F_x = m \left(\frac{v_f^2 - v_{0x}^2}{2\Delta x} \right) \quad \checkmark$$

Rearranging

$$F_x \Delta x = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_{0x}^2 \\ = K_f - K_i = \Delta K \quad \checkmark$$

Substituting from (1) in above

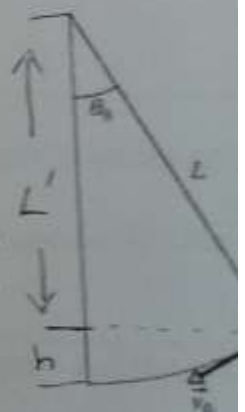
$$Fd \cos \phi = \Delta K$$

This can be written in the form of the dot product of the two vectors which by definition is the work done by the force:

$$\vec{F} \cdot \vec{d} = W = \Delta K \quad \checkmark$$

- (b) The figure shows a thin rod, of length $L = 2.00$ m and negligible mass, that can pivot about one end to rotate in a vertical circle. A ball of mass $m = 5.00$ kg is attached to the other end of the rod. The rod is pulled aside to an angle $\theta_0 = 30^\circ$ and released with an initial velocity $\vec{v}_0 = 0$ m/s. As the ball descends to its lowest point:

Die figuur dui 'n dun staaf, van lengte $L = 2.00$ m en weglaatbare massa aan, wat om 'n punt aan die een kant in 'n vertikale sirkel kan roteer. 'n Bal met massa $m = 5.00$ kg word aan die ander kant vasgemaak. Die staaf word een kant toe getrek tot 'n hoek $\theta_0 = 30^\circ$ en word dan losgelaat met 'n aanvanklike snelheid $\vec{v}_0 = 0$ m/s. Soos die bal na die laagste punt toe daal:



- i. How much work does the gravitational force do on it?
Hoeveel arbeid word verrig deur die swaartekrag op die bal?

Determine first the height above the lowest point of the ball.

$$\cos \theta = \frac{\text{adjacent}}{\text{hyp}}$$

$$\cos 30^\circ = \frac{L'}{2}$$

$$L' = 2 \cos 30^\circ$$

$$= 1.73 \text{ m} \quad \checkmark$$

Vertical change in height of ball: $h = L - L' = 0.27 \text{ m} \quad \checkmark$

The vertical component of the displacement vector is downward, same direction as the force - so we obtain

$$W_g = \vec{F}_g \cdot \vec{d} = mgh \cos(0^\circ) = mgL(1 - \cos \theta)$$

$$= (5.00 \text{ kg})(9.80 \text{ m/s}^2)(0.27 \text{ m})$$

$$= 13.2 \text{ J.} \quad \checkmark$$

$$W = -\Delta U = -mg\Delta h \quad \checkmark$$

$$= -(5.00 \text{ kg})(9.8 \text{ m/s}^2)(-0.27 \text{ m})$$

$$= +13.2 \text{ J} \quad \checkmark$$

- ii. Calculate the change in the gravitational potential energy of the ball-Earth system.
Bereken die verandering in die gravitasie potensiele energie van die bal-Aarde sisteem.

$$\Delta U = -W_g = -mgh = -13.2 \text{ J.} \quad \checkmark \checkmark$$

- iii. Calculate the speed of the ball at its lowest point.

Bereken die spoed van die bal wanneer dit by die laagste punt is.

Let $U = 0 \text{ J}$ when ball is at the lowest point. $\checkmark$

Then the potential energy is $U = mgh$ at the position when released. Thus from conservation of mechanical energy, we have

$$K_i + U_i = K_f + U_f \quad \checkmark$$

$$0 + mgh = \frac{1}{2}mv^2 + 0$$

which leads to

$$v = \sqrt{\frac{2mgh}{m}} = \sqrt{2gh} \quad \checkmark$$

With $h = 0.27 \text{ m}$ and $\theta = 30.0^\circ$ we find $v = 2.30 \text{ m/s.} \quad \checkmark$

- iv. Does the speed in (iii) increase, decrease or remain the same if the mass is increased?
 Give reasons for your answer.

From the above equation in (iii) we see that the speed is independent of the mass $\checkmark$ and therefore the speed will be the same. $\checkmark$

Question 5 / Vraag 5

A 1400 kg car moving at 5.3 m/s is initially travelling north along the positive direction of a y-axis. After completing a 90° right-hand turn in 4.6 s, the inattentive driver drives into a tree, which stops the car in 350 ms.

'n 1400 kg motorkar beweeg teen 5.3 m/s en ry aanvanklik noordwaarts langs die positiewe rigting van 'n y-as. Die kar voltooi 'n 90° draai na regs in 4.6 s, en die daarna ry die onoplettende bestuurder in 'n boom vas, en die kar kom tot stilstand in 350 ms.

- i. In unit-vector notation, what is the impulse on the car due to the turn?

In eenheidsvektor notasie, wat is die impuls op die kar as gevolg van die draai?

The initial momentum of the car is

$$\vec{p}_i = m\vec{v}_i = (1400 \text{ kg})(5.3 \text{ m/s})\hat{j} = (7400 \text{ kg} \cdot \text{m/s})\hat{j} \quad \checkmark$$

and the final momentum after making the turn is

$$\vec{p}_f = (7400 \text{ kg} \cdot \text{m/s})\hat{i} \quad \checkmark$$

(note that the magnitude remains the same, only the direction is changed). Thus, the impulse is

$$\vec{J} = \vec{p}_f - \vec{p}_i = (7.4 \times 10^3 \text{ N} \cdot \text{s})\hat{i} - (7.4 \times 10^3 \text{ N} \cdot \text{s})\hat{j}$$

- ii. What is the magnitude and direction of the average force that acts on the car during

- turn?

Wat is die grootte en rigting van die gemiddelde krag wat inwerk op die motor gedurende die draai?

The average force on the car during the turn is

$$F_{\text{avg}} = \frac{\vec{J}}{\Delta t} = \frac{\vec{p}_f - \vec{p}_i}{\Delta t} = \frac{(7.4 \times 10^3 \text{ N} \cdot \text{s})\hat{i} - (7.4 \times 10^3 \text{ N} \cdot \text{s})\hat{j}}{4.6 \text{ s}} \quad \checkmark$$

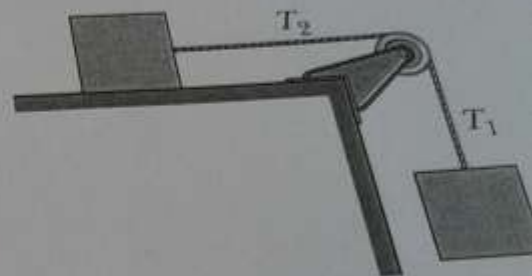
$$= 1600 \text{ N} \hat{i} - 1600 \text{ N} \hat{j} \quad \checkmark$$

and its magnitude is $F_{\text{avg}} = (1600 \text{ N})\sqrt{2} = 2.3 \times 10^3 \text{ N} \quad \checkmark$

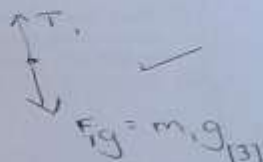
Directed at 45° below the positive x axis.

Question 6 / Vraag 6

The figure shows two 6.20 kg blocks that are connected by a massless string over a pulley of radius 2.40 cm and rotational inertia $7.40 \times 10^{-4} \text{ kg} \cdot \text{m}^2$. The string does not slip on the pulley. It is not known whether there is friction between the table and the sliding block. The pulley's axis is frictionless. When the system is released from rest, the pulley turns through 0.650 rad in 91.0 ms and the acceleration of the blocks is constant. Draw all the free body diagrams.



Die figuur dui twee 6.20 kg blokke aan. Hulle word verbind deur 'n massalose tou wat oor 'n katrol met radius 2.40 cm en traagheidsmoment van $7.40 \times 10^{-4} \text{ kg} \cdot \text{m}^2$, loop. Die tou gly nie oor die katrol nie. Dit is nie bekend of daar wrywing tussen die tafel en die glyende blok is nie. Die katrol se as is wrywingloos. Wanneer die sisteem uit rus los gelaat word, draai die katrol deur 0.650 rad in 91.0 ms en die versnelling van die blokke is konstant. Skets al die vry-liggaam diagramme.



- i. Calculate the magnitude of the pulley's angular acceleration.
Bereken die grootte van die katrol se hoekversnelling

[3]

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2 \Rightarrow \alpha = \frac{2\theta}{t^2} = \frac{2(0.65 \text{ rad})}{(0.0910 \text{ s})^2} = 157 \text{ rad/s}^2 \quad \checkmark \checkmark$$

- ii. Calculate the magnitude of either block's acceleration.
Bereken die versnelling van enige van die blokke.

[3]

$a = R\alpha$ - we obtain

$$a = R\alpha = (0.024 \text{ m}) 157 \text{ rad/s}^2 = 3.77 \text{ m/s}^2 \quad \checkmark \checkmark$$

- iii. Calculate the tension T_1 .
Bereken die spankrag T_1 ?

From the free body diagram for the hanging block:

$$\sum F_{\text{hanging}} = -ma = T_1 - mg \quad \checkmark$$

$$T_1 = m(g - a) = 6.20(9.8 - 3.77) \quad \checkmark$$

$$= 37.4 \text{ N} \quad \checkmark$$

- iv. Calculate the tension T_2 .
Bereken die spankrag T_2 ?

- iv. Calculate the tension T_2 .
Bereken die spankrag T_2 ?

[3]

From the free body diagram for the pulley:

$$\sum \tau = -I\alpha = -T_2 R + T_1 R \quad \checkmark$$

$$T_2 = \frac{-I\alpha + T_1 R}{R} = \frac{(-7.40 \times 10^{-4}) (\alpha - \text{value from (i)}) + (\text{Value from (iii)}) (2.4 \times 10^{-2})}{(2.4 \times 10^{-2})} \quad \checkmark$$

$$= 32.6 \text{ N} \quad \checkmark$$

Question 7 / Vraag 7

An oscillator consists of a block of mass 0.500 kg connected to a spring. When oscillating with an amplitude of 35.0 cm, the oscillator repeats its motion every 0.350 s. Determine the following:
 'n Ossillator bestaan uit 'n blok met massa 0.500 kg wat vas is aan 'n veer. Wanneer die sisteem ossilleer met 'n amplitude van 35.0 cm, herhaal die ossillator se beweging elke 0.350 s. Bepaal die volgende:

- i. The period.
 Die periode.

$$T = 0.350 \text{ s}$$

[2]

- ii. The angular frequency.
 Die hoek frekwensie.

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.350} = 17.95 \text{ rad/s}$$

[3]

- iii. The spring constant.
 Die veerkonstante.

$$\omega = \sqrt{\frac{k}{m}}$$

$$k = \omega^2 m = 161 \text{ N/m}$$

[3]

- iv. The magnitude of the maximum force on the block due to the spring.
 Die grootte van die maksimum krag op die blok as gevolg van die veer.

$$a_m = \omega^2 x_m$$

$$= (\omega \text{ from ii})^2 (0.35 \text{ m})$$

$$a_m = \omega^2 x_m$$

$$= (\omega \text{ from ii})^2 (0.35 \text{ m})$$

$$= 112.8 \text{ m/s}^2$$

$$F_{\max} = ma_m$$

$$= 0.50 \text{ kg} (112.8 \text{ m/s}^2)$$

$$= 56.4 \text{ N}$$

OR

$$F_{\max} = |-kx_m|$$

$$= 161 \text{ N/m} (0.35 \text{ m})$$

$$= 56.4 \text{ N}$$

- v. Write down the general equation for the position of the block as a function of time.
 Skryf die algemene vergelyking vir die posisie van die blok as 'n funksie van tyd neer.

$$x = x_m \cos(\omega t + \phi)$$

$$x = 0.35 \cos(17.95t + \phi) \text{ cm}$$

$$\text{If at } t = 0, x = x_m \text{ then } \phi = 0$$

[3]

Enjoy the rest of your day / Geniet die res van jou dag